DATA ANALYTICS | PROJECT DOCUMENTATION

Customer Behaviour Analysis

End-to-end analysis: Python | PostgreSQL | Power BI

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Document Control

Field	Detail
Project	Customer Behaviour Analysis
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Version / Date	1.0 / 29 October 2026
Data source	customer_shopping_behavior.csv (3,900 records, 18 attributes)
Tools	Python (pandas, SQLAlchemy), PostgreSQL, Power BI, Jupyter Notebook
Status	Analysis and dashboard complete; remediation items listed in Section 12

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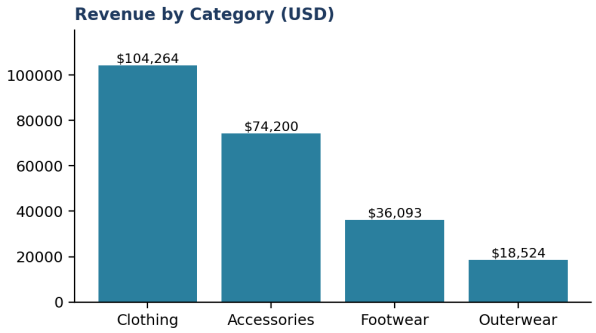
Appendix A. Final Schema

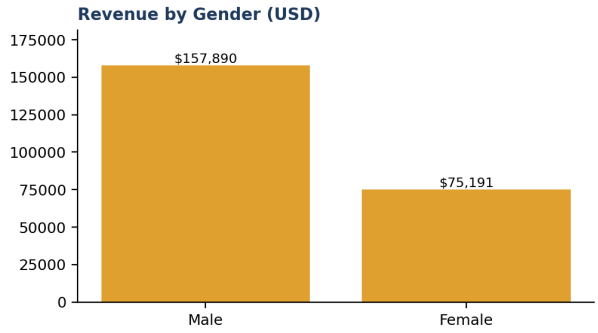
1. Executive Summary

This project examines how customers of a fashion retail business shop: who they are, what they buy, how they pay and receive orders, and which levers (discounts, subscriptions, shipping) correlate with spending. The work was delivered in four stages: data cleaning and feature engineering in Python, loading of the clean dataset into PostgreSQL, ten business questions answered in SQL, and an interactive Power BI dashboard for stakeholders.

Key Findings

Metric	Result
Customers analysed	3,900
Total revenue	$233,081
Average order value	$59.76
Average review rating	3.75 / 5
Revenue share, male vs female	68% vs 32% ($157,890 vs $75,191)
Largest category	Clothing: $104,264 (45% of orders)
Subscribed customers	27% (1,053); average spend $59.49 vs $59.87 for non-subscribers
Purchases with a discount	43% (1,677 of 3,900)
Loyal customers (more than 10 purchases)	80% (3,116)





2. Project Objective and Scope

Objective. Understand customer purchasing behaviour in order to support decisions on marketing, discounting, subscription strategy, shipping options and product focus.

In scope. Data cleaning, feature engineering, SQL-based analysis and dashboard development.

Out of scope. Predictive modelling, clustering and time-series analysis (the dataset contains no transaction dates).

3. Deliverables and Artefacts

Artefact	Purpose
customer_shopping_behavior.csv	Raw source dataset, 3,900 rows by 18 columns
Ananlysis_of_Customer_s_Behaviour.ipynb	Cleaning, feature engineering and PostgreSQL load
Customer_Behavior_Analysis_SQL_query_and_insights.sql	Ten business-question queries
Customer_Behaviour_Analysis_Dashboard.pbix	Interactive Power BI dashboard
Untitled.ipynb	Empty notebook, no content (candidate for deletion)

4. Methodology and Workflow

Phase	Activity	Tool
1	Load and profile raw data	Python / pandas
2	Clean data: missing values, naming, redundant columns	Python / pandas
3	Engineer features: age group, purchase frequency	Python / pandas
4	Load clean table into database	SQLAlchemy, PostgreSQL
5	Answer business questions	PostgreSQL (SQL)
6	Visualise and share insights	Power BI

5. Data Understanding

The CSV was loaded into a pandas DataFrame and profiled with `head()`, `info()` and `describe(include='all')`.

Dataset Structure

- **Volume:** 3,900 rows and 18 columns.
- **Numeric:** Customer ID, Age, Purchase Amount (USD), Review Rating, Previous Purchases.
- **Categorical:** Gender, Item Purchased, Category, Location, Size, Color, Season, Subscription Status, Shipping Type, Discount Applied, Promo Code Used, Payment Method, Frequency of Purchases.

Profiling Summary

Attribute	Observation
Age	18 to 70, mean 44.07
Purchase amount	Mean $59.76, range $20 to $100
Review rating	Mean 3.75, range 2.5 to 5.0; 37 missing values
Unique values	25 products, 4 categories, 50 locations, 25 colours
Most frequent	Blouse (171 orders), Clothing (1,737 orders), Montana (96), Olive, Spring
Gender	2,652 male, 1,248 female

6. Data Cleaning

6.1 Missing Values

A null check showed that only **Review Rating** contained missing values: **37 of 3,900 records (0.9%)**. Each gap was filled with the **mean rating of the record's own category**, which reflects category-specific rating behaviour better than a single global mean.

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.mean()))
```

Validation: a repeated null check returned 0 missing values across all columns.

6.2 Column Name Standardisation

Column names were converted to lower case with underscores, and the revenue column was shortened, making every field safe to reference in SQL.

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ','_')
df = df.rename(columns={'purchase_amount_(usd)':'purchase_amount'})
```

6.3 Removal of a Redundant Column

Equality of *Discount Applied* and *Promo Code Used* was tested before removal:

```
(df['discount_applied'] == df['promo_code_used']).all()   # Output: True
```

The columns were identical on every row, so `promo_code_used` was dropped to avoid duplicated information.

7. Feature Engineering

7.1 Age Groups

Age was divided into four equal-sized groups with quartile binning (`pd.qcut`), which keeps group sizes balanced for fair comparison.

Age group	Age range	Customers
Young Adult	18 to 31	1,028
Adult	32 to 44	942
Middle-Age	45 to 57	986
Senior	58 to 70	944

7.2 Purchase Frequency in Days

Text frequencies were mapped to a numeric number of days for use in calculations.

```
frequecy_mapping = {'Fortnightly':14,'Weekly':7,'Monthly':30,'Quaterly':90,
                    'Annually':365,'Every 3 months':90}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequecy_mapping)
```

> **Finding:** The mapping keys do not match the data values. The dataset uses *Quarterly*, *Every 3 Months* (capital M) and *Bi-Weekly*. Those three groups (563 + 584 + 547 = 1,694 rows, about 43%) therefore become NaN. **Fix:** use the keys 'Quarterly', 'Every 3 Months' and add 'Bi-Weekly': 14, then reload the table.

8. Database Loading (PostgreSQL)

The cleaned DataFrame was written to a local PostgreSQL database using SQLAlchemy and psycopg2.

Setting	Value
Database	Customers_behaviour
Table	customer
Host / Port	localhost / 5432
Method	df.to_sql(..., if_exists="replace", index=False)

Output: `Data successfully loaded into table 'customer' in database 'Customers_behaviour'.`

> **Security note:** the notebook stores the database password in plain text. Replace it with an environment variable (for example `os.getenv("PG_PASSWORD")`) before sharing the notebook.

9. SQL Analysis

Ten business questions were answered in PostgreSQL. The source SQL file contains queries only, so the results below were reproduced from the same dataset using identical logic.

Q1. Revenue by gender

```
select gender, SUM(purchase_amount) as revenue
from customer
group by gender;
```

Gender	Revenue (USD)
Male	157,890
Female	75,191

> **Insight:** Male customers drive about 68% of revenue, partly because they are 68% of the customer base (2,652 vs 1,248). Average spend per customer is the fairer comparison.

Q2. Discount users who still spent above average

```
select customer_id, purchase_amount
from customer
where discount_applied = 'Yes'
  and purchase_amount >= (select AVG(purchase_amount) from customer);
```

Output: 839 customers (about 50% of 1,677 discounted purchases) spent at or above the $59.76 average.

> **Insight:** Discounts do not necessarily mean smaller baskets; half of discount users still spend above average.

Q3. Top 5 products by average review rating

```
select item_purchased,
       ROUND(AVG(review_rating::numeric),2) as "Average Product Rating"
from customer
group by item_purchased
order by avg(review_rating) desc
limit 5;
```

Rank	Product	Avg rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Handbag	3.78

> **Insight:** Ratings are tightly clustered (3.78 to 3.86), so differences between the top products are small.

Q4. Average purchase: Standard vs Express shipping

```
select shipping_type, ROUND(AVG(purchase_amount),2)
from customer
where shipping_type in ('Standard','Express')
group by shipping_type;
```

Shipping type	Avg purchase (USD)
Express	60.48
Standard	58.46

Insight: Express customers spend about $2 more per order, a mild signal only.

Q5. Do subscribers spend more? (corrected query)

```
select subscription_status,
       COUNT(customer_id)            as total_customers,
       ROUND(AVG(purchase_amount),2) as "Average Spend",
       SUM(purchase_amount)          as total_revenue
from customer
group by subscription_status
order by total_revenue;
```

Subscribed	Customers	Avg spend (USD)	Revenue (USD)
No	2,847	59.87	170,436
Yes	1,053	59.49	62,645

Insight: Subscribers do not spend more per order. Subscription acts as a loyalty lever rather than a spend lever, and only 27% of customers use it.

Correction: the original Q5 had a misplaced parenthesis in COUNT(), an incomplete ROUND(AVG(...)) expression, "groub by", and the misspelt column "subscribtion_status". The version above is the corrected query.

Q6. Top 5 products by share of discounted purchases

```
select item_purchased,
  ROUND(100 * SUM(CASE WHEN discount_applied='Yes' THEN 1 ELSE 0 END)/COUNT(*),2)
    as discount_rate
from customer
group by item_purchased
order by discount_rate desc
limit 5;
```

Rank	Product	Discount rate (%)
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

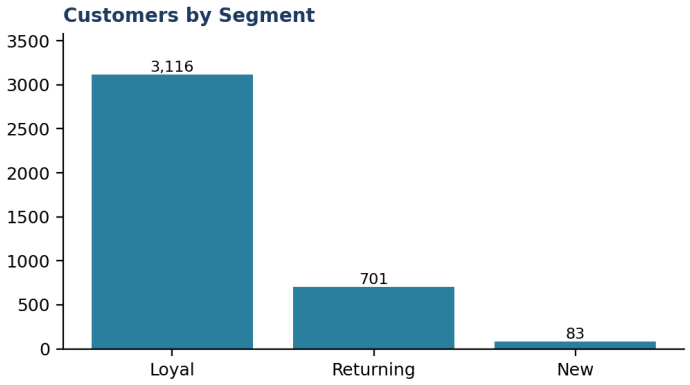
Insight: Nearly half of sales for these products are discounted, which makes them the first candidates for a margin review. Use 100.0 instead of 100 to avoid integer truncation.

Q7. Customer segmentation by purchase history

```
with customer_type as (
  select customer_id, previous_purchases,
  CASE WHEN previous_purchases = 1 THEN 'New'
       WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
       ELSE 'Loyal Customers' END as customer_segment
  from customer)
select customer_segment, count(*) as "Number of Customers"
from customer_type group by customer_segment;
```

Segment	Customers	Share
Loyal (more than 10)	3,116	79.9%
Returning (2 to 10)	701	18.0%
New (1)	83	2.1%

Insight: The base is overwhelmingly loyal while new customers are scarce, which points to an acquisition gap rather than a retention problem.



Q8. Top 3 products within each category

```
with item_counts as (
  select category, item_purchased,
  COUNT(customer_id) as total_orders,
  ROW_NUMBER() over(partition by category order by count(customer_id) DESC) as item_rank
  from customer group by category, item_purchased)
select item_rank, category, item_purchased, total_orders
from item_counts where item_rank <= 3;
```

Category	#1	#2	#3
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Accessories	Jewelry (171)	Belt (161)	Sunglasses (161)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	n/a (two products)

Insight: Counts are very close within categories and ties exist (Blouse/Pants, Belt/Sunglasses). ROW_NUMBER breaks ties arbitrarily; RANK or DENSE_RANK would make ties explicit.

Q9. Are repeat buyers likely to subscribe?

```
select subscription_status, count(customer_id) as repeat_buyers
from customer
where previous_purchases > 5
group by subscription_status;
```

Subscribed	Repeat buyers (more than 5 purchases)
No	2,518
Yes	958

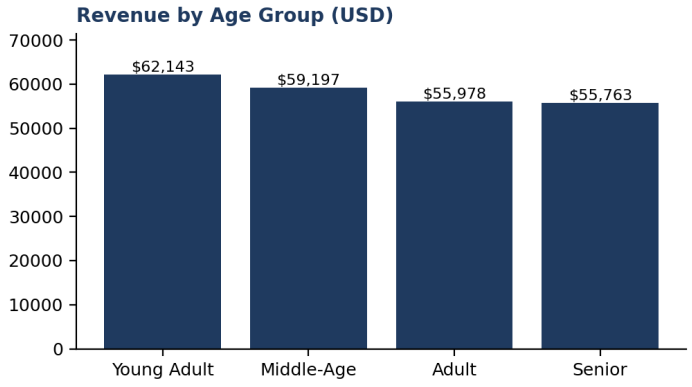
Insight: Only about 28% of repeat buyers subscribe, close to the overall 27% rate, so loyalty does not translate into subscription. This is a clear conversion opportunity.

Q10. Revenue contribution by age group

```
select age_group, SUM(purchase_amount) as total_revenue
from customer
group by age_group
order by total_revenue desc;
```

Age group	Revenue (USD)
Young Adult	62,143
Middle-Age	59,197
Adult	55,978
Senior	55,763

Insight: Revenue is well balanced; the top group is only about 11% above the lowest.



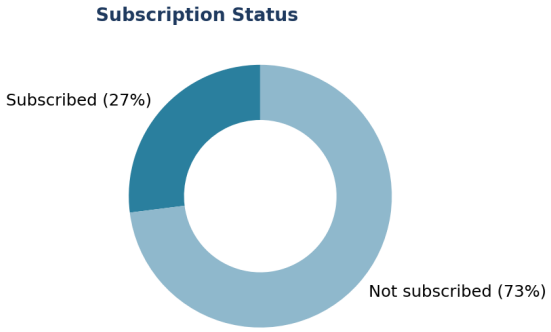
10. Power BI Dashboard

The dashboard connects to the PostgreSQL `customer` table and is built as a single page (3000 x 1500 canvas). Four slicers filter all visuals simultaneously.

Component	Type	Purpose
KPI cards	Card	Number of Customers, Average Purchase Amount, Average Review Rating (DAX measures)
Customers Subscription status %	Donut chart	Subscribed vs non-subscribed share
Revenue by Category	Column chart	Total purchase amount per category
Sales by Category	Column chart	Customer volume per category
Revenue by Age Group	Bar chart	Total purchase amount per age group
Sales by Age Group	Bar chart	Customer volume per age group
Slicers	Filter	Subscription status, Gender, Category, Shipping type

Review Findings

- The two "Sales by ..." charts sum `customer_id`, which adds ID numbers and is not meaningful. Replace with a distinct count of customers.
- The average purchase card uses the rupee symbol while the data is in USD. Align the format to dollars.
- Consider adding a page title, data-source note, and visuals for discount usage and shipping type, which are analysed in SQL but not yet shown.



11. Business Recommendations

#	Recommendation	Supporting evidence
1	Grow subscriptions with a tangible benefit and measure retention rather than spend	27% subscribed; subscribers spend the same; only 28% of repeat buyers subscribe
2	Invest in customer acquisition	New customers are only 2.1% of the base
3	Review discount dependency	Hat, Sneakers, Coat, Sweater, Pants discounted on about half of sales
4	Study the female customer segment	32% of customers and revenue; spend and category preferences are unexplored
5	Prioritise Clothing and Accessories	Together about 76% of revenue

12. Known Issues and Next Steps

#	Item	Priority
1	Fix frequency mapping spelling and case so purchase_frequency_days has no NaN; reload table	High
2	Correct SQL Q5 syntax and the subscribtion_status spelling in the .sql file	High
3	Replace sum of customer_id with a count in two Power BI charts	High
4	Remove plain-text database password from the notebook	High
5	Standardise currency to USD in Power BI	Medium
6	Add EDA visuals (distributions, correlation heatmap) to the notebook	Medium
7	Delete the empty Untitled.ipynb; add a README and dashboard screenshots	Low
8	Optional extension: customer segmentation (K-Means) or loyalty prediction	Optional

Appendix A. Final Schema (table: customer)

Column	Type	Description
customer_id	int	Unique customer identifier
age	int	Customer age (18 to 70)
gender	text	Male / Female
item_purchased	text	One of 25 products
category	text	Clothing, Accessories, Footwear, Outerwear
purchase_amount	int	Order value in USD
location	text	US state (50 values)
size, color	text	Product size and colour
season	text	Spring, Summer, Fall, Winter
review_rating	float	2.5 to 5.0; missing values filled by category mean
subscription_status	text	Yes / No
shipping_type	text	Standard, Express, Next Day Air, 2-Day, Store Pickup, Free Shipping
discount_applied	text	Yes / No
previous_purchases	int	Earlier purchases (1 to 50)
payment_method	text	Payment mode
frequency_of_purchases	text	Purchase cadence label
age_group	category	Engineered: Young Adult, Adult, Middle-Age, Senior
purchase_frequency_days	float	Engineered; contains NaN until mapping fix (Section 7.2)